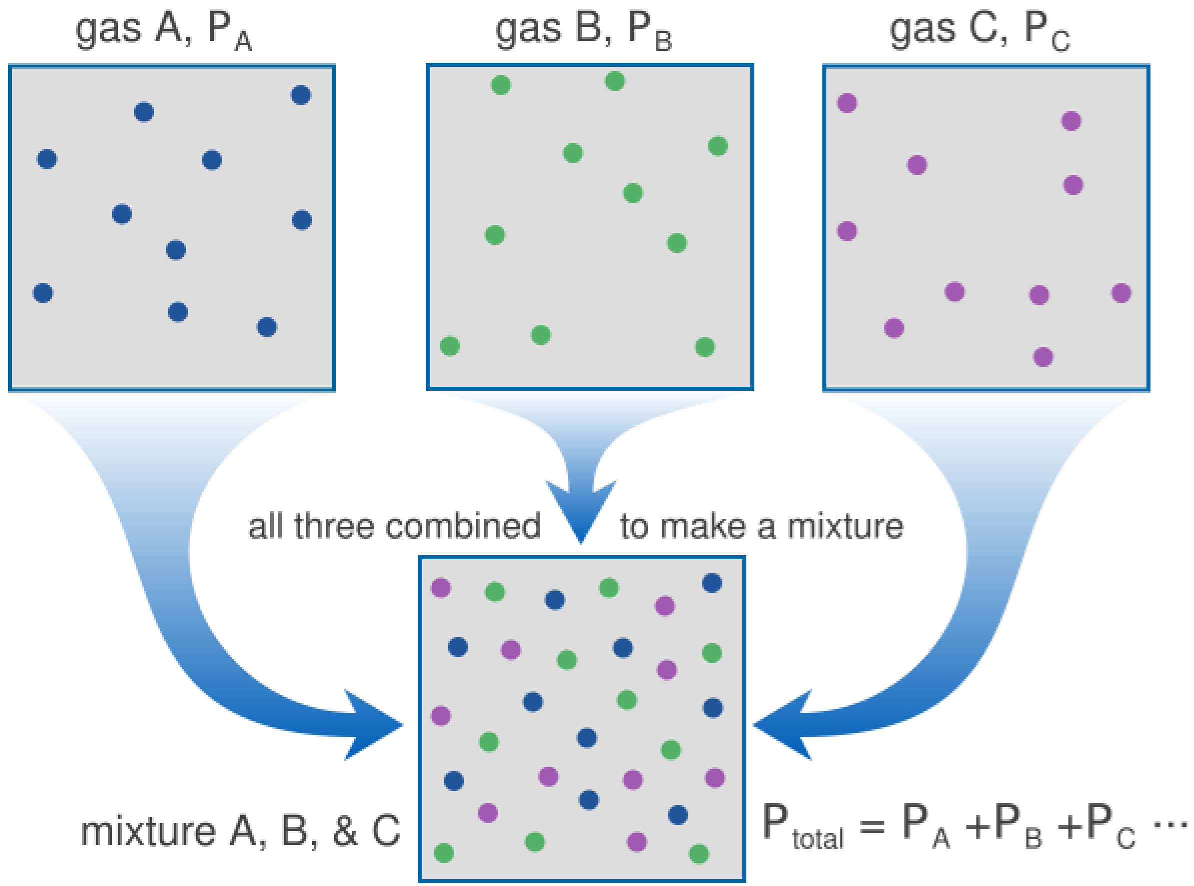
Physical Chemistry (I)

Lecture 17

Gas and Liquid Mixtures

Gas Mixture



(Image: http://ch301.cm.utexas.edu/gases/index.php#mixtures/partial-pressure.html)

Dalton's Law

Partial pressure for species J is

$$p_J = x_J p$$

so

$$p_\mathrm{A} + p_\mathrm{B} + \cdots = (x_\mathrm{A} + x_\mathrm{B} + \cdots)p = p$$

Valid for all gases (not only perfect)

Chemical Potential

For a perfect gas,

$$G_{\mathrm{m}}(p) = G_{\mathrm{m}}^{\ominus} + RT \ln \frac{p}{p^{\ominus}}$$

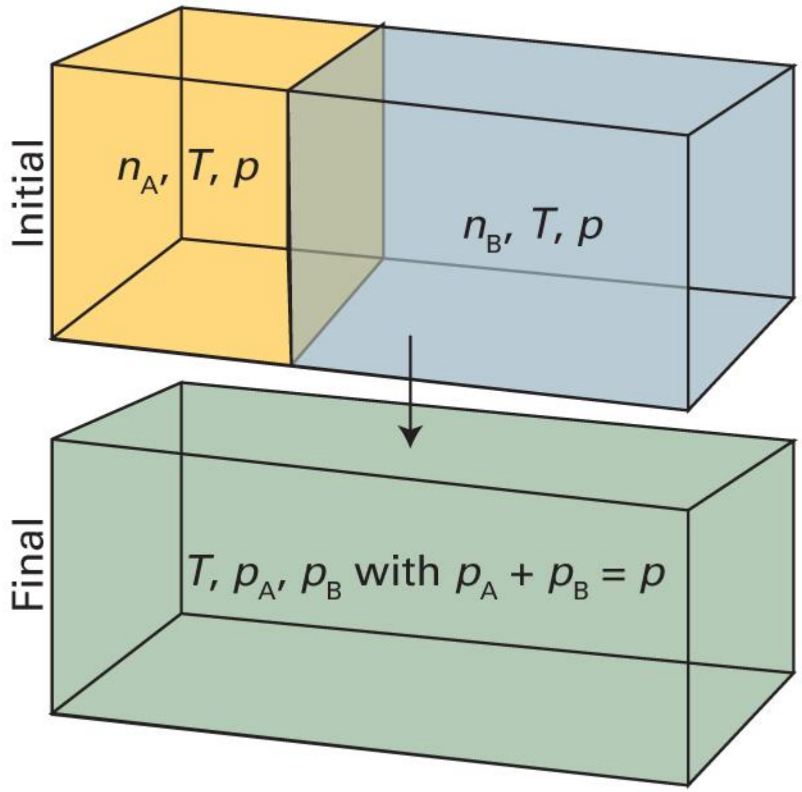
or,

$$\mu = \mu^{\ominus} + RT \ln \frac{p}{p^{\ominus}}$$

where $\mu^{\ominus}$ is the standard chemical potential (at 1 bar).

Mixing of Gases

(at constant T and p)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.6)

Total Gibbs Energy

$$G = G_{\mathrm{A}} + G_{\mathrm{B}} = n_{\mathrm{A}}\mu_{\mathrm{A}} + n_{\mathrm{B}}\mu_{\mathrm{B}}$$

$$\mu = \mu^{\ominus} + RT\ln(p/p^{\ominus})$$

- Initial state

$$G_i = n_{\mathrm{A}}\{\mu_{\mathrm{A}}^{\ominus} + RT\ln(p/p^{\ominus})\} + n_{\mathrm{B}}\{\mu_{\mathrm{B}}^{\ominus} + RT\ln(p/p^{\ominus})\}$$

- Final state

$$G_f = n_{\mathrm{A}}\{\mu_{\mathrm{A}}^{\ominus} + RT\ln(p_{\mathrm{A}}/p^{\ominus})\} + n_{\mathrm{B}}\{\mu_{\mathrm{B}}^{\ominus} + RT\ln(p_{\mathrm{B}}/p^{\ominus})\}$$

Topic 5A

Gibbs Energy of Mixing

$$\Delta_{\mathrm{mix}}G = G_f - G_i = n_{\mathrm{A}}RT\ln\left(\frac{p_{\mathrm{A}}}{p}\right) + n_{\mathrm{B}}RT\ln\left(\frac{p_{\mathrm{B}}}{p}\right)$$

Using Dalton's law,

$$\Delta_{\mathrm{mix}}G = n_{\mathrm{A}}RT\ln x_{\mathrm{A}} + n_{\mathrm{B}}RT\ln x_{\mathrm{B}}$$

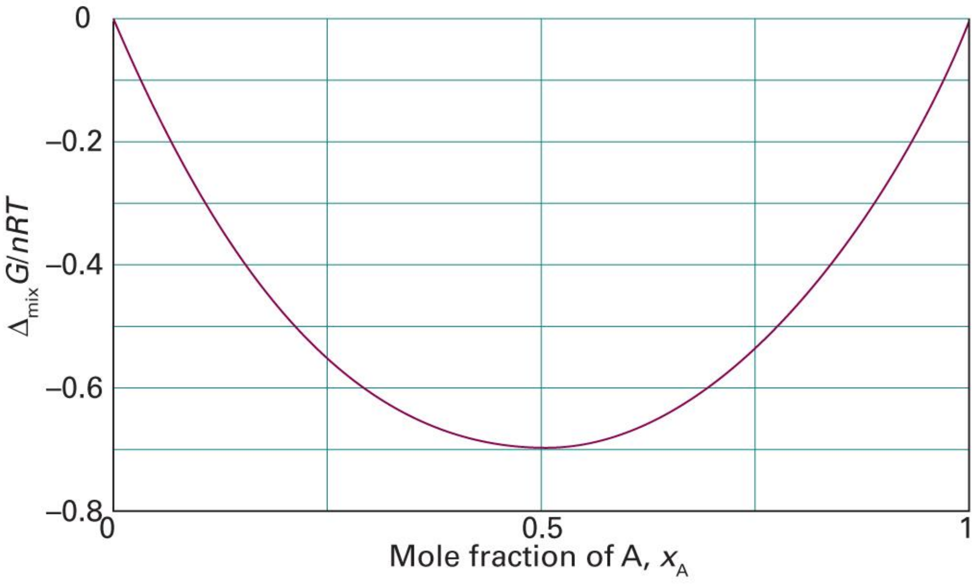
and

$$\Delta_{\mathrm{mix}}G = nRT(x_{\mathrm{A}}\ln x_{\mathrm{A}} + x_{\mathrm{B}}\ln x_{\mathrm{B}})$$

Note: since $\ln x < 0$ for $x < 1$, $\Delta_{\mathrm{mix}}G < 0$ always.

Gibbs Energy of Mixing

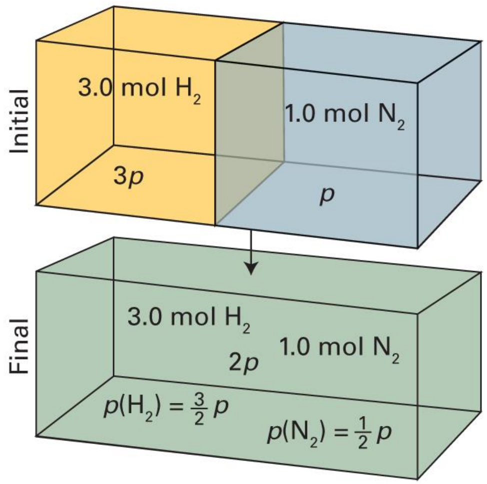
$$\Delta_{\text{mix}}G = nRT\{x_{\text{A}} \ln x_{\text{A}} + (1 - x_{\text{A}}) \ln(1 - x_{\text{A}})\}$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.7)

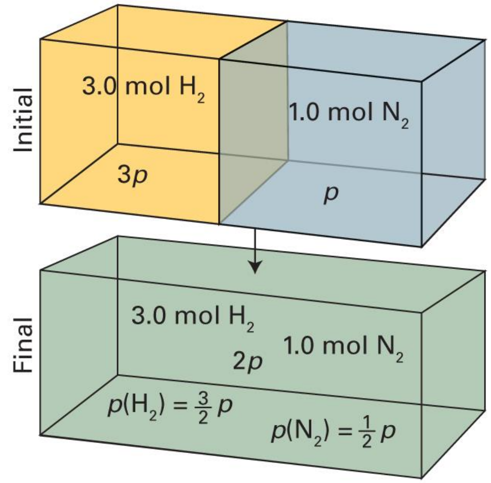
Example 5A.3

A container is divided into two equal compartments. One contains 3.0 mol $H_2(g)$ at 25°C; the other contains 1.0 mol $N_2(g)$ at 25°C. Calculate the Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.7)

Example 5A.3



- H_2: amount 3.0 mol, volume $V \rightarrow 2V$, pressure $3p \rightarrow (3/2)p$
- N_2: amount 1.0 mol, volume $V \rightarrow 2V$, pressure $p \rightarrow (1/2)p$

$$G = G_\mathrm{A} + G_\mathrm{B} = n_\mathrm{A}\mu_\mathrm{A} + n_\mathrm{B}\mu_\mathrm{B}$$

$$\mu = \mu^\ominus + RT\ln(p/p^\ominus)$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.7)

Example 5A.3

$$G = G_{\mathrm{A}} + G_{\mathrm{B}} = n_{\mathrm{A}}\mu_{\mathrm{A}} + n_{\mathrm{B}}\mu_{\mathrm{B}}$$

$$\mu = \mu^{\ominus} + RT\ln(p/p^{\ominus})$$

Total Gibbs energies are

$$G_i = n_{\mathrm{A}}\{\mu_{\mathrm{A}}^{\ominus} + RT\ln(3p/p^{\ominus})\} + n_{\mathrm{B}}\{\mu_{\mathrm{B}}^{\ominus} + RT\ln(p/p^{\ominus})\}$$

$$G_f = n_{\mathrm{A}}\{\mu_{\mathrm{A}}^{\ominus} + RT\ln(3p/2p^{\ominus})\} + n_{\mathrm{B}}\{\mu_{\mathrm{B}}^{\ominus} + RT\ln(p/2p^{\ominus})\}$$

Therefore,

$$\Delta_{\mathrm{mix}}G = G_f - G_i = -n_{\mathrm{A}}RT\ln 2 - n_{\mathrm{B}}RT\ln 2 = -(n_{\mathrm{A}} + n_{\mathrm{B}})RT\ln 2$$

$= -(3.00\text{ mol}+1.00\text{ mol})\times(8.3145\text{ J K}^{-1}\text{ mol}^{-1})\times(298\text{ K})\times 0.693$

$= -6.9\text{ kJ}$

Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

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Real Gases

For a perfect gas,

$$\mu = \mu^{\ominus} + RT \ln \frac{p}{p^{\ominus}}$$

For a real gas,

$$\mu = \mu^{\ominus} + RT \ln \frac{f}{p^{\ominus}}$$

where f is the fugacity.

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Properties of Fugacity

- As $p \to 0$, $f \to p$
- Fugacity coefficient $\phi = f/p$
- Molecular interpretation
  - $f > p$ or $\phi > 1$: repulsion-dominant
  - $f = p$ or $\phi = 1$: perfect-gas-like
  - $f < p$ or $\phi < 1$: attraction-dominant

NOTE

$$\mu = \mu^\ominus + \int_{p^\ominus}^{p} V_\mathrm{m}\,\mathrm{d}p$$

$$= \mu^\ominus + \int_{p^\ominus}^{p} \frac{RT}{p}\,\mathrm{d}p$$

$$= \mu^\ominus + RT\ln\frac{p}{p^\ominus}$$

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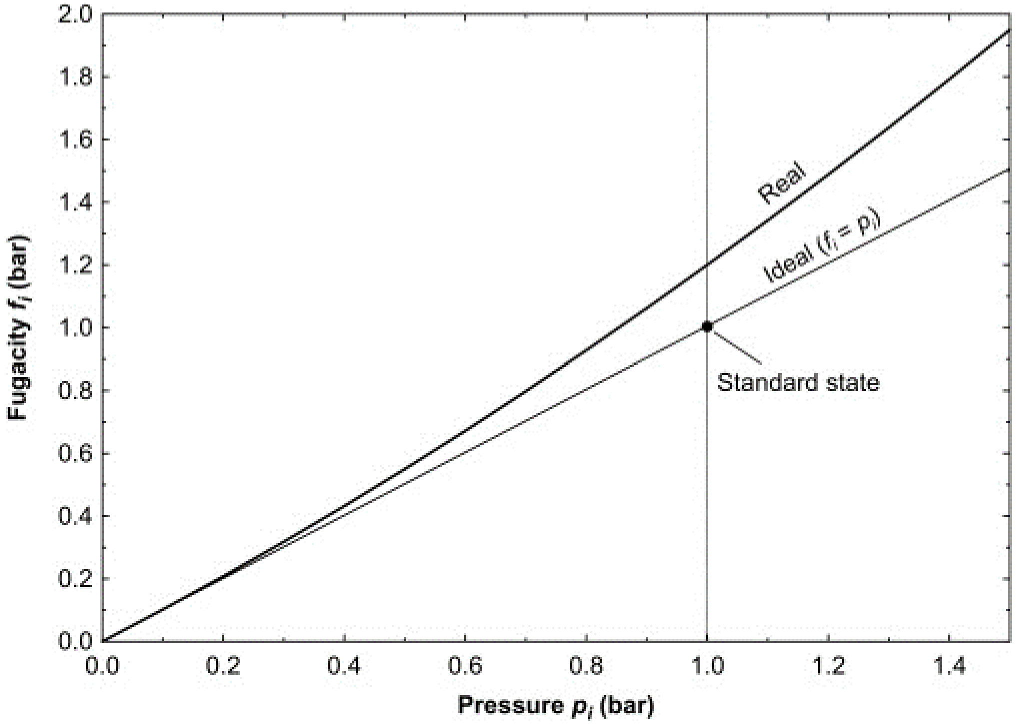
Fugacity and Pressure



(Image: https://doi.org/10.1016/B978-0-12-801494-3.00002-6)

Mixing of Liquids

$$\mu_A(l) = \mu_A(g)$$

When A is a pure substance,

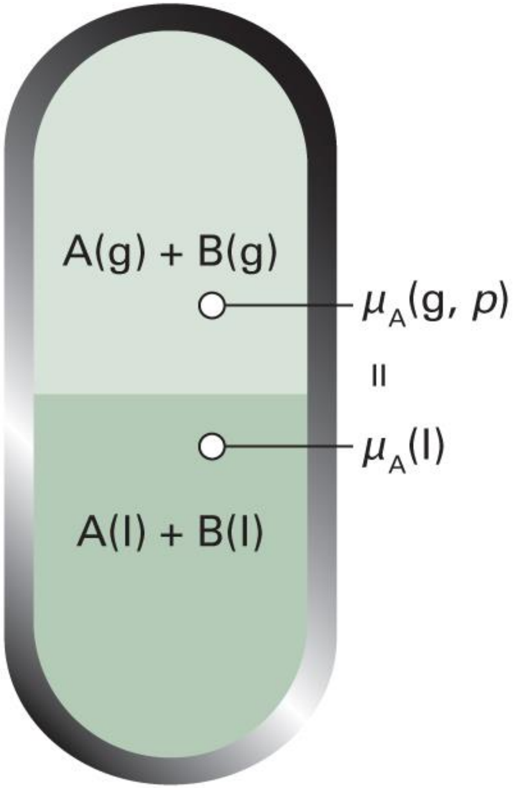
$$\underbrace{\mu_A^*(l)}_{\text{liquid}} = \underbrace{\mu_A^\ominus(g) + RT\ln(p_A^*/p^\ominus)}_{\text{(perfect) gas}}$$

If A is mixed with another substance B,

$$\mu_A(l) = \mu_A^\ominus(g) + RT\ln(p_A/p^\ominus)$$

Subtraction leads to

$$\mu_A(l) = \mu_A^*(l) + RT\ln(p_A/p_A^*)$$

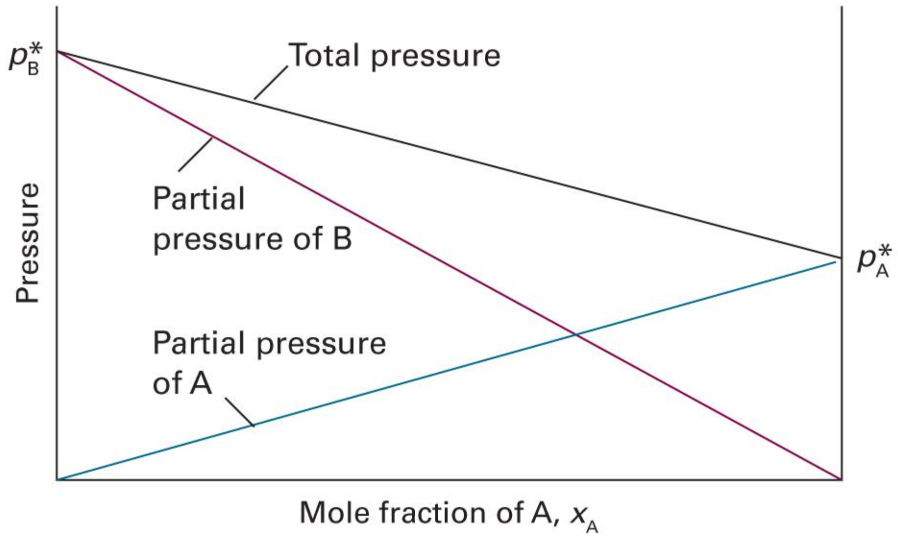


(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.10)

Raoult's Law: Ideal Solution

$$p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$$

- Ideal solutions: mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.11)

Chemical Potential

$$\mu_A(l) = \mu_A^*(l) + RT\ln(p_A/p_A^*)$$

For an ideal solution,

$$p_A = x_A p_A^*$$

Hence,

$$\mu_A(l) = \mu_A^*(l) + RT\ln x_A$$

As $x_A < 1$, $\mu_A(l) < \mu_A^*(l)$ and mixed phase is more stable

Ideal Solutions

- Before mixing:

$$G_i = n_\mathrm{A}\mu_\mathrm{A}^* + n_\mathrm{B}\mu_\mathrm{B}^*$$

- After mixing:

$$G_f = n_\mathrm{A}\mu_\mathrm{A} + n_\mathrm{B}\mu_\mathrm{B}$$

$$= n_\mathrm{A}(\mu_\mathrm{A}^* + RT\ln x_\mathrm{A}) + n_\mathrm{B}(\mu_\mathrm{B}^* + RT\ln x_\mathrm{B})$$

- Gibbs energy of mixing

$$\Delta_\mathrm{mix}G = G_f - G_i = n_\mathrm{A}RT\ln x_\mathrm{A} + n_\mathrm{B}RT\ln x_\mathrm{B}$$

$$\Delta_\mathrm{mix}G = nRT(x_\mathrm{A}\ln x_\mathrm{A} + x_\mathrm{B}\ln x_\mathrm{B})$$

Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

- Entropy of mixing

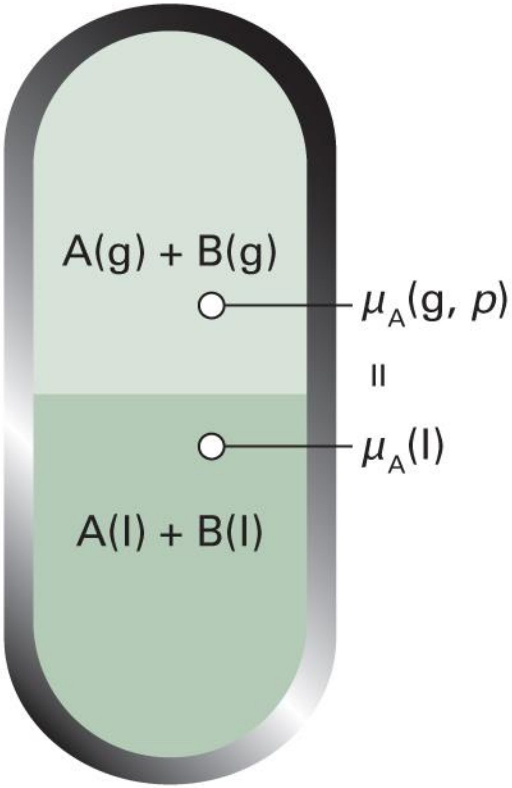
$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

(Same with perfect gas mixtures)

Liquid-Vapor Equilibrium



- Constraints

$$\mu_{\mathrm{A}}(g) = \mu_{\mathrm{A}}(l)$$

$$\mu_{\mathrm{B}}(g) = \mu_{\mathrm{B}}(l)$$

- May have $x_{\mathrm{A}}(g) \neq x_{\mathrm{A}}(l)$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.10)

Brief Illustration 5A.3

Consider a mixture of benzene and methylbenzene, which form an approximately ideal solution. The vapour pressure of pure benzene at 20°C is 75.0 Torr and that of pure methylbenzene is 21.0 Torr at the same temperature. What are the mole fractions in the vapour when the two liquids are in an equimolar mixture?

In an equimolar mixture,

$$x_{\text{benzene}}(l) = x_{\text{methylbenzene}}(l) = 0.5$$

Thus,

$$p_{\text{benzene}} = 75.0\text{ Torr} \times 0.5 = 37.5\text{ Torr}$$

$$p_{\text{methylbenzene}} = 21.0\text{ Torr} \times 0.5 = 10.5\text{ Torr}$$

Brief Illustration 5A.3

The total pressure is 37.5 Torr + 10.5 Torr = 48.0 Torr.

From Dalton's law,

$x_{\mathrm{benzene}}(g)$ = (37.5 Torr)/(48.0 Torr) = 0.78

$x_{\mathrm{methylbenzene}}(g)$ = (10.5 Torr)/(48.0 Torr) = 0.22

Vapor Pressures

From Raoult's law,

$$p_\mathrm{A} = x_\mathrm{A} p_\mathrm{A}^*, \ p_\mathrm{B} = x_\mathrm{B} p_\mathrm{B}^*$$

The total vapor pressure is

$$p = p_\mathrm{A} + p_\mathrm{B} = x_\mathrm{A} p_\mathrm{A}^* + x_\mathrm{B} p_\mathrm{B}^* = x_\mathrm{A} p_\mathrm{A}^* + (1 - x_\mathrm{A}) p_\mathrm{B}^*$$

which becomes

$$p = p_\mathrm{B}^* + (p_\mathrm{A}^* - p_\mathrm{B}^*) x_\mathrm{A}$$

Vapor Compositions

$$p = p_{\mathrm{B}}^{*} + (p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*})x_{\mathrm{A}}$$

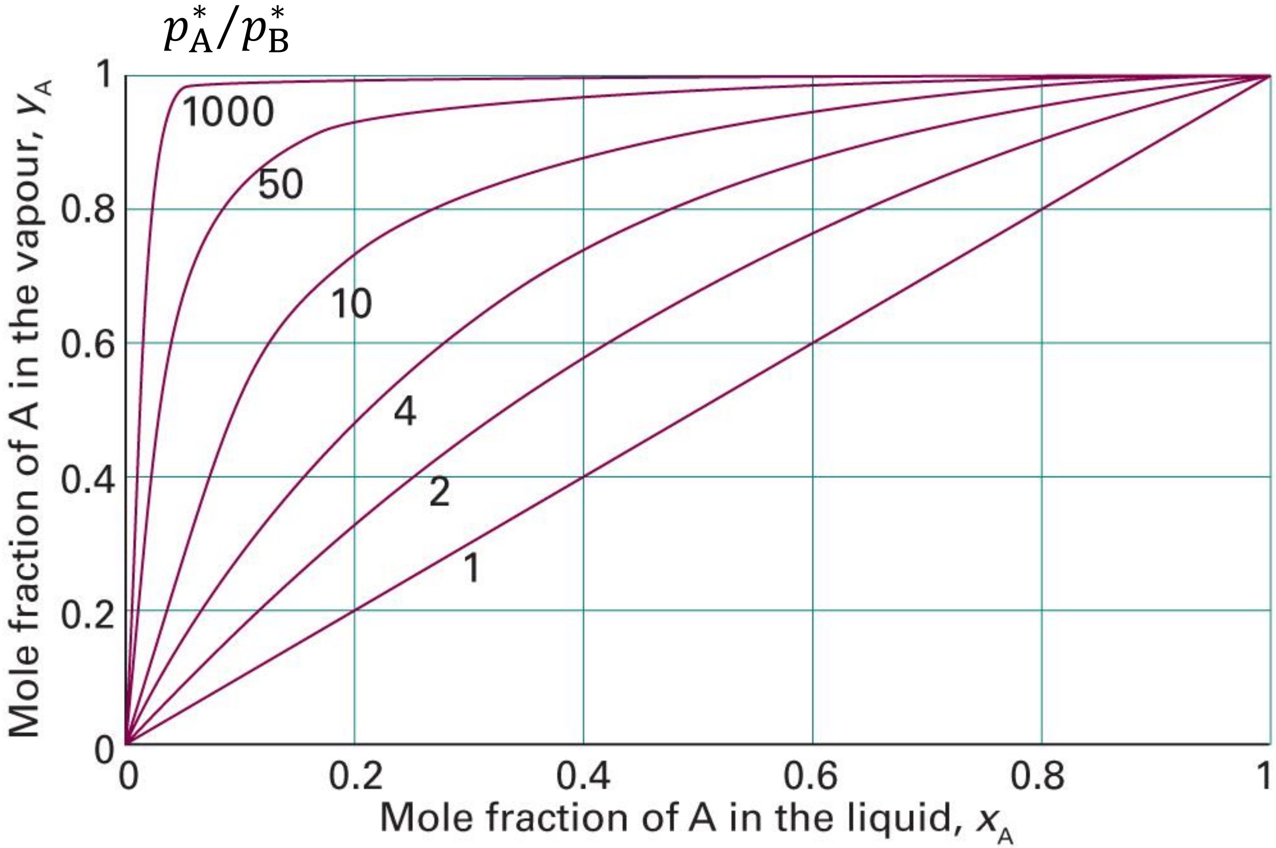
Denote the mole fractions in the vapor as y_{A} and y_{B}:

$$y_{\mathrm{A}} = p_{\mathrm{A}}/p\,, \;\; y_{\mathrm{B}} = p_{\mathrm{B}}/p$$

Now,

$$y_{\mathrm{A}} = \frac{x_{\mathrm{A}}p_{\mathrm{A}}^{*}}{p_{\mathrm{B}}^{*} + (p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*})x_{\mathrm{A}}}, \qquad y_{\mathrm{B}} = 1 - y_{\mathrm{A}}$$

Vapor Compositions



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5C.2)

x_{A} as a Function of y_{A}

$$y_{\mathrm{A}} = \frac{x_{\mathrm{A}} p_{\mathrm{A}}^{*}}{p_{\mathrm{B}}^{*} + (p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*}) x_{\mathrm{A}}}$$

$$p_{\mathrm{B}}^{*} y_{\mathrm{A}} + (p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*}) x_{\mathrm{A}} y_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$$

$$p_{\mathrm{B}}^{*} y_{\mathrm{A}} = x_{\mathrm{A}} \{ p_{\mathrm{A}}^{*} - (p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*}) y_{\mathrm{A}} \}$$

Hence,

$$x_{\mathrm{A}} = \frac{y_{\mathrm{A}} p_{\mathrm{B}}^{*}}{p_{\mathrm{A}}^{*} + (p_{\mathrm{B}}^{*} - p_{\mathrm{A}}^{*}) y_{\mathrm{A}}}$$

Total Vapor Pressure

$$x_{\mathrm{A}} = \frac{y_{\mathrm{A}} p_{\mathrm{B}}^{*}}{p_{\mathrm{A}}^{*} + (p_{\mathrm{B}}^{*} - p_{\mathrm{A}}^{*}) y_{\mathrm{A}}}$$

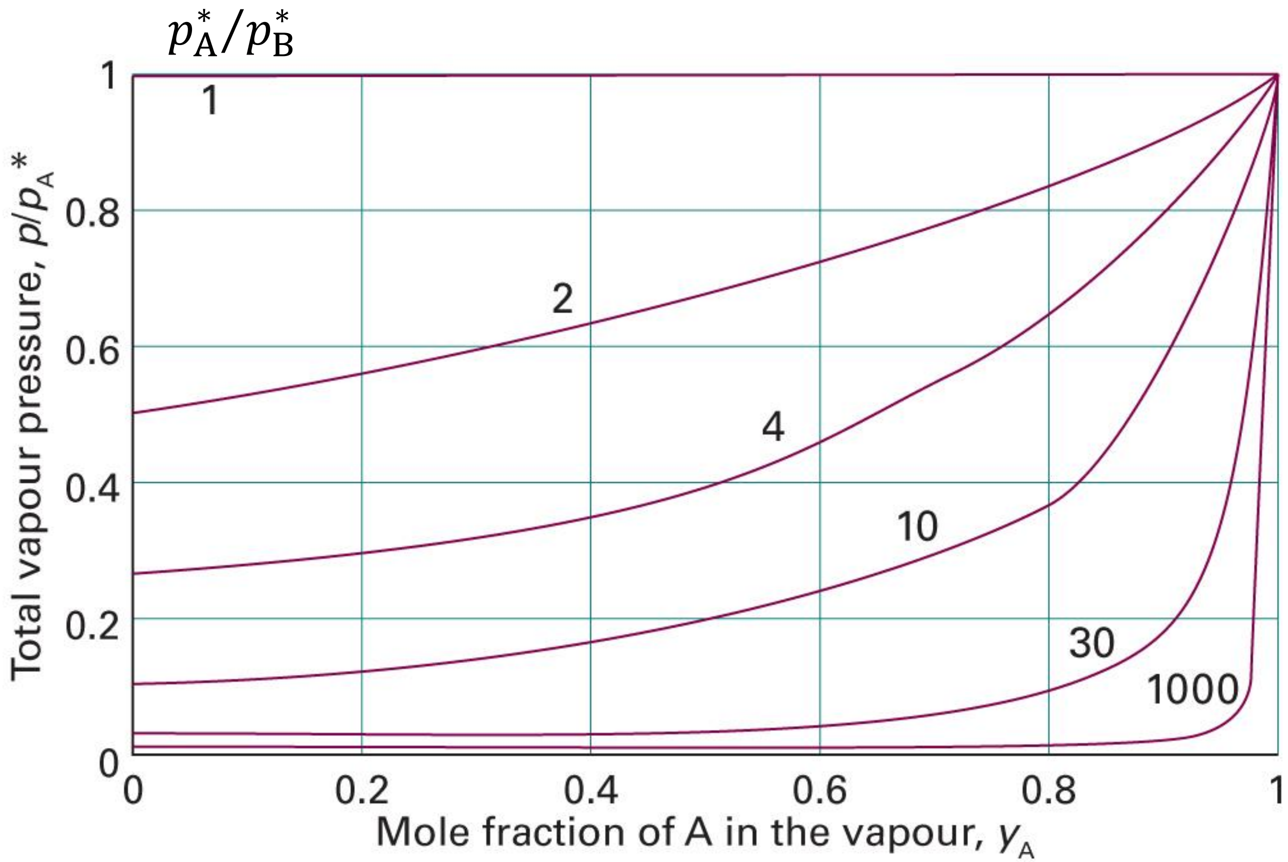
The total vapor pressure is

$$p = p_{\mathrm{B}}^{*} + (p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*}) x_{\mathrm{A}} = p_{\mathrm{B}}^{*} + \frac{(p_{\mathrm{A}}^{*} - p_{\mathrm{B}}^{*}) y_{\mathrm{A}} p_{\mathrm{B}}^{*}}{p_{\mathrm{A}}^{*} + (p_{\mathrm{B}}^{*} - p_{\mathrm{A}}^{*}) y_{\mathrm{A}}}$$

which leads to

$$p = \frac{p_{\mathrm{A}}^{*} p_{\mathrm{B}}^{*}}{p_{\mathrm{A}}^{*} + (p_{\mathrm{B}}^{*} - p_{\mathrm{A}}^{*}) y_{\mathrm{A}}}$$

Total Vapor Pressure



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 5C.3)

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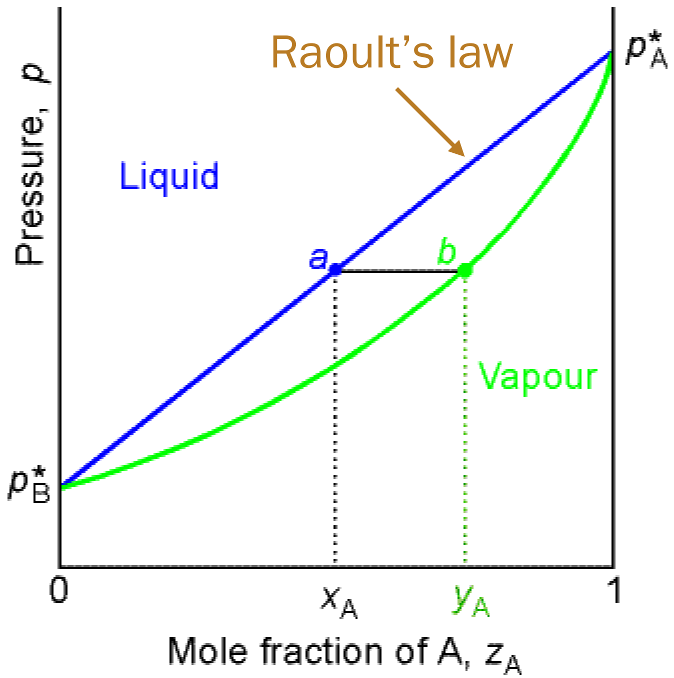
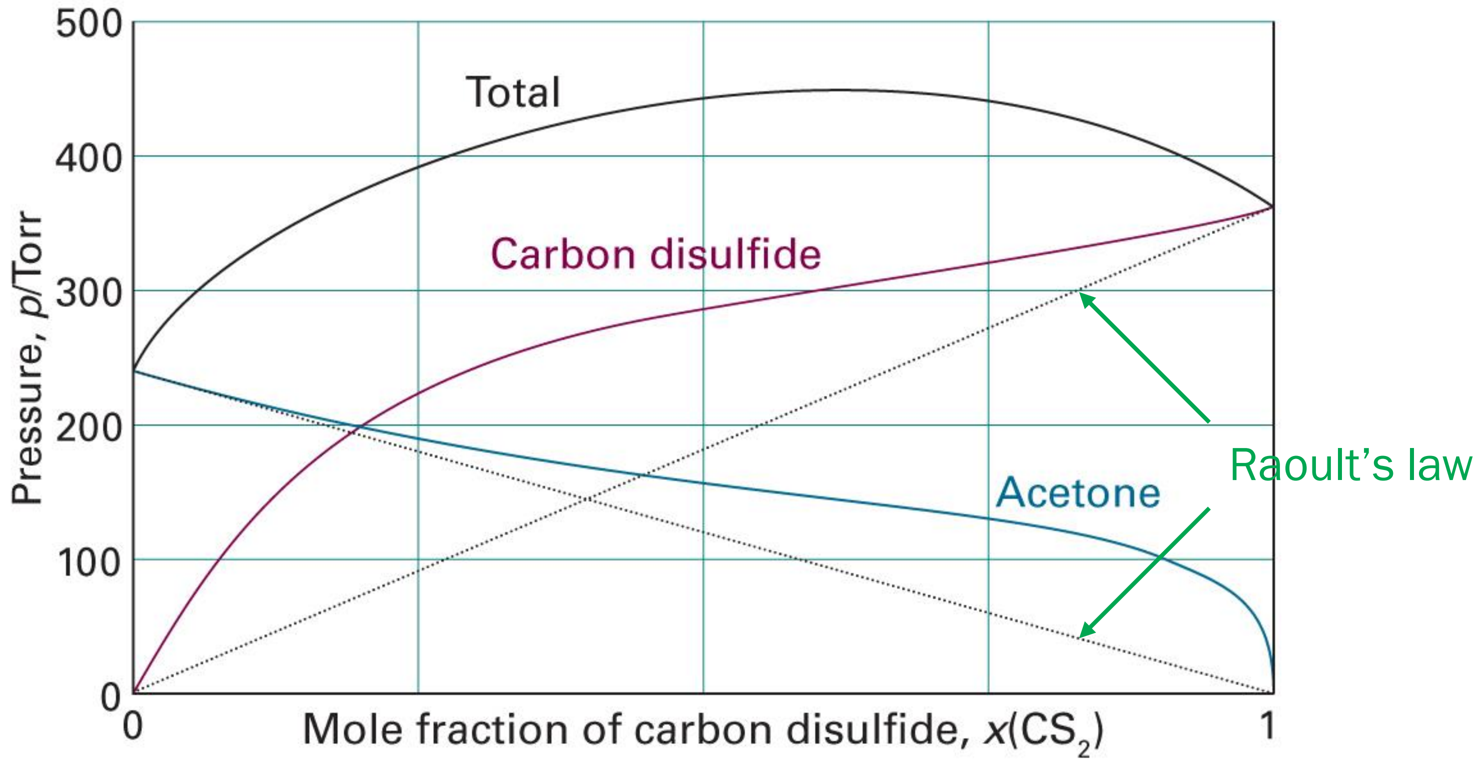
Pressure-Composition Diagram



(Image: http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm)

Real Solutions: Deviations



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.12)

Henry's Law

$$p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$$

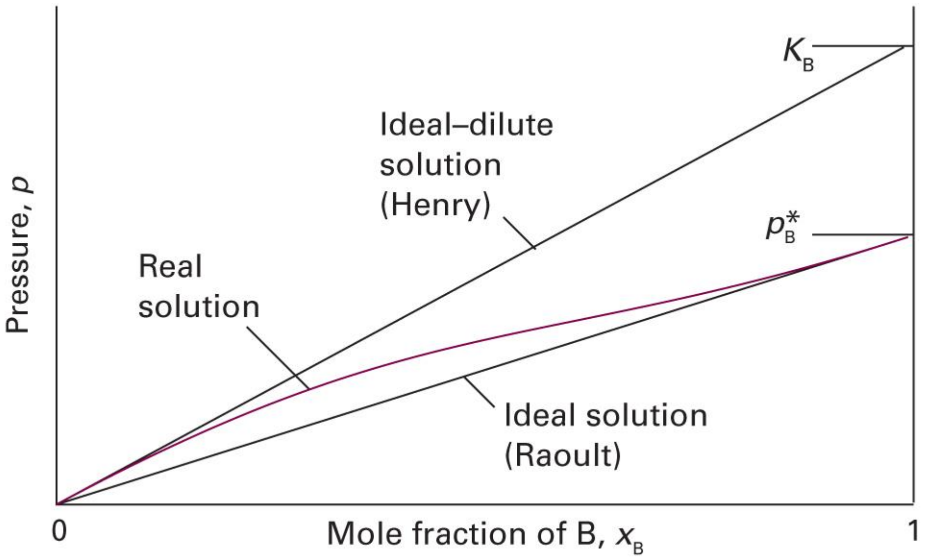
- Ideal-dilute solutions: mixtures for which the solute B obeys Henry's law and the solvent A obeys Raoult's law

The molality, b, is often instead used:

$$p_B = b_{\mathrm{B}} K_{\mathrm{B}}$$

Limiting Laws for Real Solutions

- Raoult's law: reliable as $x \to 1$
- Henry's law: reliable as $x \to 0$



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 5A.13)

Chemical Potentials

- Solvent ($x_{\mathrm{A}} \to 1$): Raoult's law

$$\mu_{\mathrm{A}}(l) = \mu_{\mathrm{A}}^{*}(l) + RT\ln(p_{\mathrm{A}}/p_{\mathrm{A}}^{*}) = \mu_{\mathrm{A}}^{*}(l) + RT\ln x_{\mathrm{A}}$$

- Solute ($x_{\mathrm{B}} \to 0$): Henry's law

$$\mu_{\mathrm{B}}(l) = \mu_{\mathrm{B}}^{*}(l) + RT\ln(p_{\mathrm{B}}/p_{\mathrm{B}}^{*}) = \mu_{\mathrm{B}}^{*}(l) + RT\ln(K_{\mathrm{B}}x_{\mathrm{B}}/p_{\mathrm{B}}^{*})$$

$$= \underbrace{\mu_{\mathrm{B}}^{*}(l) + RT\ln(K_{\mathrm{B}}/p_{\mathrm{B}}^{*})}_{\mu_{\mathrm{B}}^{\ominus}(l)} + RT\ln x_{\mathrm{B}}$$

$$= \mu_{\mathrm{B}}^{\ominus}(l) + RT\ln x_{\mathrm{B}}$$

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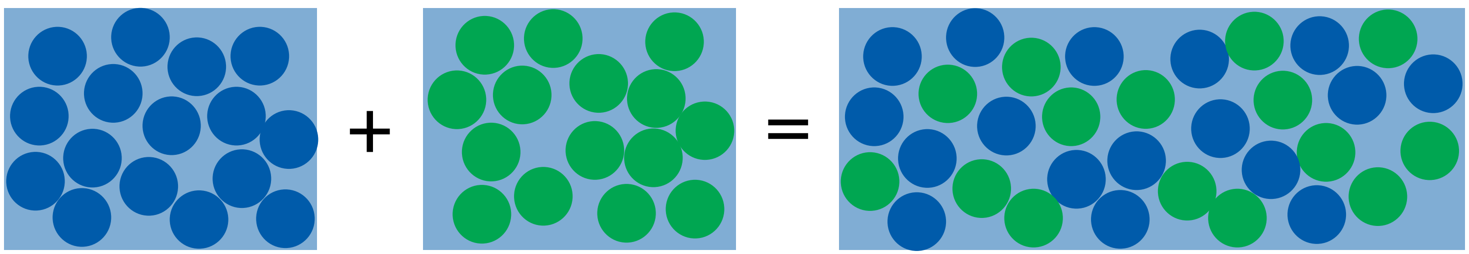
Microscopic Model

Pure substances Mixture



	Pure substance		Mixture
Number of A molecules	N_A	0	N_A
Number of B molecules	0	N_B	N_B
Total number of molecules	N_A	N_B	$N_A + N_B$

Calculate the chemical potentials

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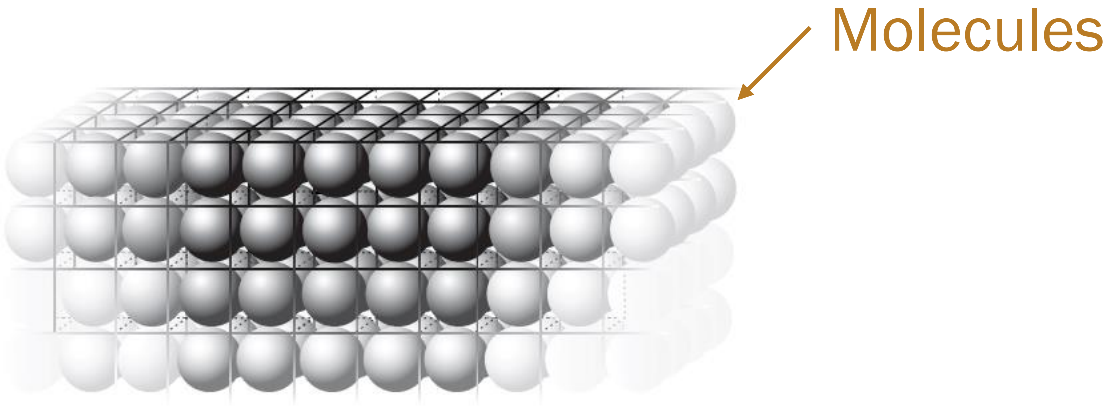
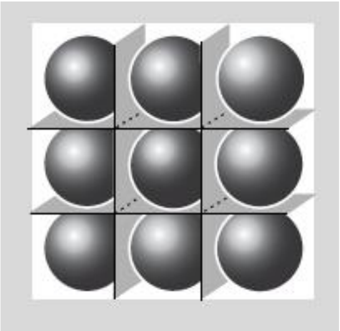
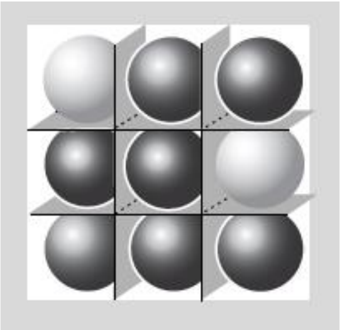
Lattice Model



Pure substance



Mixture



(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figures 14.1 and 16.2)

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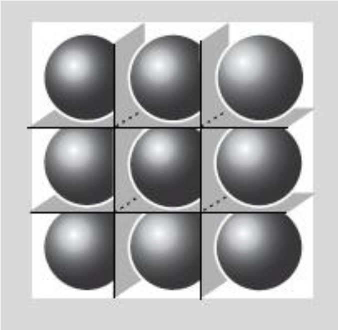
Numbers of Microstates

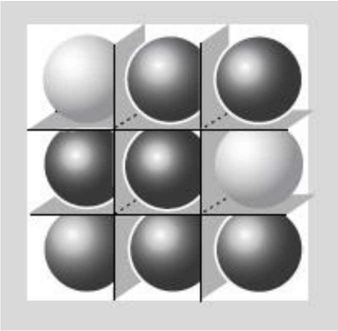
Pure



- Number of rooms: N
- Number of molecules: N

$$W = 1$$

Mixed



- Number of rooms: $N_\mathrm{A} + N_\mathrm{B}$
- Number of A molecules: N_A
- Number of B molecules: N_B

$$W = {}_{N_\mathrm{A}+N_\mathrm{B}}C_{N_\mathrm{A}} = (N_\mathrm{A} + N_\mathrm{B})!/(N_\mathrm{A}!\,N_\mathrm{B}!)$$

(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figure 16.2)

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Entropy of Mixing

- Pure A: $S_\mathrm{A} = k_\mathrm{B} \ln W_\mathrm{A} = k_\mathrm{B} \ln 1 = 0$
- Pure B: $S_\mathrm{B} = k_\mathrm{B} \ln W_\mathrm{B} = k_\mathrm{B} \ln 1 = 0$
- Mixture:

$$S_\mathrm{A+B} = k_\mathrm{B} \ln W_\mathrm{A+B} = k_\mathrm{B} \ln \frac{(N_\mathrm{A} + N_\mathrm{B})!}{N_\mathrm{A}!\, N_\mathrm{B}!}$$

- Entropy of mixing

$$\Delta_\mathrm{mix} S = S_\mathrm{A+B} - S_\mathrm{A} - S_\mathrm{B} = k_\mathrm{B} \ln \frac{(N_\mathrm{A} + N_\mathrm{B})!}{N_\mathrm{A}!\, N_\mathrm{B}!}$$

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Entropy of Mixing

$$\Delta_{\text{mix}}S = k_{\text{B}} \ln \frac{(N_{\text{A}} + N_{\text{B}})!}{N_{\text{A}}!\, N_{\text{B}}!}$$

Applying Stirling's approximation,

$$\begin{aligned}
\Delta_{\text{mix}}S &= k_{\text{B}}\{\ln(N_{\text{A}} + N_{\text{B}})! - \ln N_{\text{A}}! - \ln N_{\text{B}}!\} \\
&= k_{\text{B}}\{(N_{\text{A}} + N_{\text{B}}) \ln(N_{\text{A}} + N_{\text{B}}) - N_{\text{A}} \ln N_{\text{A}} - N_{\text{B}} \ln N_{\text{B}}\} \\
&= -k_{\text{B}}\left(N_{\text{A}} \ln \frac{N_{\text{A}}}{N_{\text{A}}+N_{\text{B}}} + N_{\text{B}} \ln \frac{N_{\text{B}}}{N_{\text{A}}+N_{\text{B}}}\right) \\
&= -Nk_{\text{B}}(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}}) \\
&= -nR(x_{\text{A}} \ln x_{\text{A}} + x_{\text{B}} \ln x_{\text{B}})\text{: ideal solution entropy!}
\end{aligned}$$

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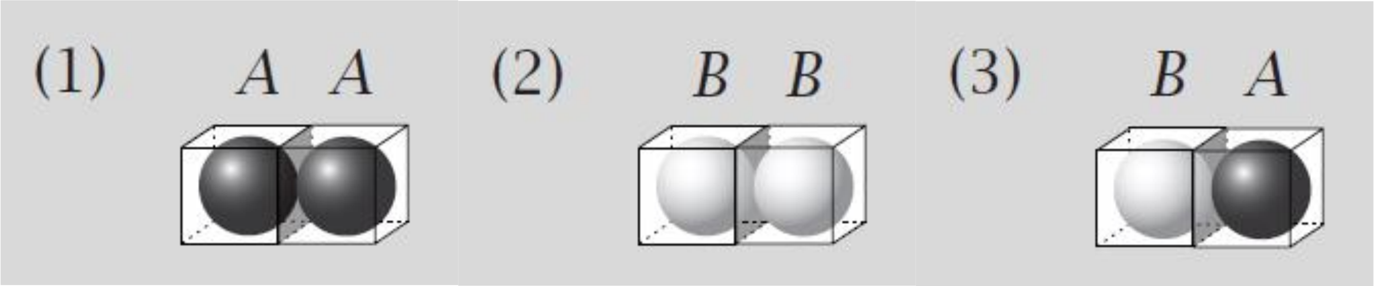
Energy of Mixture



Considering three types of contacts,

$$U = m_{\mathrm{AA}} w_{\mathrm{AA}} + m_{\mathrm{BB}} w_{\mathrm{BB}} + m_{\mathrm{AB}} w_{\mathrm{AB}}$$

where m_{ij} is the number of i-j contacts, and

w_{ij} is the energy of each i-j contact

(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figure 15.4)

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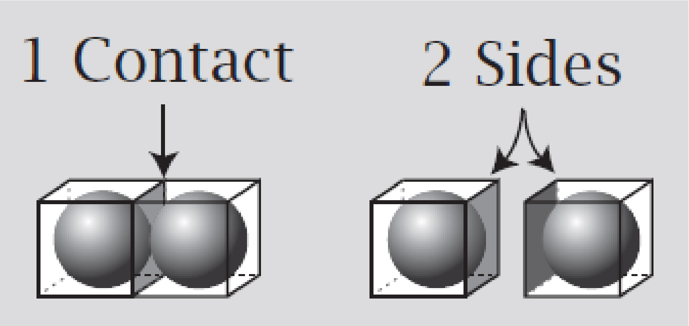
Contacts and Sides



- Each contact contributes 2 sides
- Each molecule contributes z sides
- Total number of “A” sides = $zN_\mathrm{A} = 2m_\mathrm{AA} + m_\mathrm{AB}$
- Total number of “B” sides = $zN_\mathrm{B} = 2m_\mathrm{BB} + m_\mathrm{AB}$

(Image: Dill and Bromberg, ***Molecular Driving Forces***, 2nd Ed., Figure 15.5)

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Energy of Mixture

$$zN_{\mathrm{A}} = 2m_{\mathrm{AA}} + m_{\mathrm{AB}}, \qquad zN_{\mathrm{B}} = 2m_{\mathrm{BB}} + m_{\mathrm{AB}}$$

Rearrangement leads to

$$m_{\mathrm{AA}} = (zN_{\mathrm{A}} - m_{\mathrm{AB}})/2$$

$$m_{\mathrm{BB}} = (zN_{\mathrm{B}} - m_{\mathrm{AB}})/2$$

which gives

$$U = \frac{zN_{\mathrm{A}} - m_{\mathrm{AB}}}{2} w_{\mathrm{AA}} + \frac{zN_{\mathrm{B}} - m_{\mathrm{AB}}}{2} w_{\mathrm{BB}} + m_{\mathrm{AB}} w_{\mathrm{AB}}$$

$$U = \frac{zw_{\mathrm{AA}}}{2} N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2} N_{\mathrm{B}} + \left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right) m_{\mathrm{AB}}$$

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What is $\boldsymbol{m}_{\mathbf{AB}}$?

The probability to have B in any site:

$$p_{\mathrm{B}} = N_{\mathrm{B}}/N$$

The average number of AB contacts that each A makes:

$$zp_{\mathrm{B}} = zN_{\mathrm{B}}/N$$

The total number of AB contacts:

$$m_{\mathrm{AB}} \approx N_{\mathrm{A}} \times (zN_{\mathrm{B}}/N) = zN_{\mathrm{A}}N_{\mathrm{B}}/N$$

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Energy of Mixture

$$U = \frac{zw_{\mathrm{AA}}}{2}N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2}N_{\mathrm{B}} + \left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right)\frac{zN_{\mathrm{A}}N_{\mathrm{B}}}{N}$$

$$U = \frac{zw_{\mathrm{AA}}}{2}N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2}N_{\mathrm{B}} + \xi k_{\mathrm{B}}T\frac{N_{\mathrm{A}}N_{\mathrm{B}}}{N}$$

where the exchange parameter ξ is

$$\xi = \frac{z}{k_{\mathrm{B}}T}\left(w_{\mathrm{AB}} - \frac{w_{\mathrm{AA}} + w_{\mathrm{BB}}}{2}\right)$$

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Energy of Mixing

$$U = \frac{zw_{\mathrm{AA}}}{2} N_{\mathrm{A}} + \frac{zw_{\mathrm{BB}}}{2} N_{\mathrm{B}} + \xi k_{\mathrm{B}} T \frac{N_{\mathrm{A}} N_{\mathrm{B}}}{N}$$

- Pure A: $N_{\mathrm{B}} = 0$

$$U_{\mathrm{A}} = \frac{zw_{\mathrm{AA}}}{2} N_{\mathrm{A}}$$

- Pure B: $N_{\mathrm{A}} = 0$

$$U_{\mathrm{B}} = \frac{zw_{\mathrm{BB}}}{2} N_{\mathrm{B}}$$

$$\Delta_{\mathrm{mix}} U = U_{\mathrm{A+B}} - U_{\mathrm{A}} - U_{\mathrm{B}} = \xi k_{\mathrm{B}} T \frac{N_{\mathrm{A}} N_{\mathrm{B}}}{N}$$

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Helmholtz Energy of Mixing

$$\Delta_{\mathrm{mix}}S = -nR(x_\mathrm{A}\ln x_\mathrm{A} + x_\mathrm{B}\ln x_\mathrm{B})$$

$$\Delta_{\mathrm{mix}}U = \xi k_\mathrm{B}T\frac{N_\mathrm{A}N_\mathrm{B}}{N} = \xi N k_\mathrm{B}T x_\mathrm{A}x_\mathrm{B} = \xi nRT x_\mathrm{A}x_\mathrm{B}$$

$$\Delta_{\mathrm{mix}}A = \Delta_{\mathrm{mix}}U - T\Delta_{\mathrm{mix}}S$$

$$\Delta_{\mathrm{mix}}A = nRT(x_\mathrm{A}\ln x_\mathrm{A} + x_\mathrm{B}\ln x_\mathrm{B} + \xi x_\mathrm{A}x_\mathrm{B})$$

Regular solution: $\Delta_{\mathrm{mix}}S = \Delta_{\mathrm{mix}}S^{\mathrm{ideal}}$ and $\Delta_{\mathrm{mix}}U \neq 0$

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Energy of Mixing

For a regular solution,

$$\Delta_{\text{mix}}U = nRT\xi x_{\text{A}}x_{\text{B}}$$

$$\xi = \frac{z}{k_{\text{B}}T}\left(w_{\text{AB}} - \frac{w_{\text{AA}} + w_{\text{BB}}}{2}\right)$$

- $\xi > 0$, or $w_{\text{AB}} > (w_{\text{AA}} + w_{\text{BB}})/2$: mixing is endothermic
- $\xi < 0$, or $w_{\text{AB}} < (w_{\text{AA}} + w_{\text{BB}})/2$: mixing is exothermic
- $\xi = 0$, or $w_{\text{AB}} = (w_{\text{AA}} + w_{\text{BB}})/2$: ideal solution

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Energy to Enthalpy

For liquid-liquid mixing,

$$\Delta_{\mathrm{mix}}H = H_{\mathrm{A+B}} - H_{\mathrm{A}} - H_{\mathrm{B}}$$

$$H_{\mathrm{A+B}} = U_{\mathrm{A+B}} + (pV)_{\mathrm{A+B}}$$

$$H_{\mathrm{A}} = U_{\mathrm{A}} + (pV)_{\mathrm{A}}$$

$$H_{\mathrm{B}} = U_{\mathrm{B}} + (pV)_{\mathrm{B}}$$

Hence,

$$\Delta_{\mathrm{mix}}H = \Delta_{\mathrm{mix}}U + (pV)_{\mathrm{A+B}} - (pV)_{\mathrm{A}} - (pV)_{\mathrm{B}} \approx \Delta_{\mathrm{mix}}U$$

$$\Delta_{\mathrm{mix}}G = \Delta_{\mathrm{mix}}H - T\Delta_{\mathrm{mix}}S \approx \Delta_{\mathrm{mix}}U - T\Delta_{\mathrm{mix}}S = \Delta_{\mathrm{mix}}A$$

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Model Summary

- Ideal solution
  - Raoult's law: $p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$
  - $\Delta_{\mathrm{mix}} G = -T\Delta_{\mathrm{mix}} S = nRT(x_{\mathrm{A}} \ln x_{\mathrm{A}} + x_{\mathrm{B}} \ln x_{\mathrm{B}})$
- Ideal-dilute solution
  - Solvent – Raoult's law
  - Solute – Henry's law: $p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$
- Regular solution
  - $\Delta_{\mathrm{mix}} G = \Delta_{\mathrm{mix}} H - T\Delta_{\mathrm{mix}} S^{\mathrm{ideal}}$
  - Lattice model (Bragg-Williams model): $\Delta_{\mathrm{mix}} H = nRT\xi x_{\mathrm{A}} x_{\mathrm{B}}$